Is Extreme Learning Machine Feasible? A Theoretical Assessment (Part I)

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1 Introduction

This paper provides a comprehensive theoretical assessment of the feasibility of Extreme Learning Machine (ELM). ELM is an FNN-like learning system in which connections with output neurons are adjustable, while connections with and within hidden neurons are randomly fixed. The study addresses three fundamental questions:

1. Does ELM degrade the generalization capability of traditional feedforward neural networks (FNNs) where all connections are adjusted?
2. If not, what is the number of hidden layer nodes needed to achieve optimal generalization capability?
3. How can the weak regularity of the hidden layer output matrix be guaranteed for efficient application of the generalized inverse technique?

2 ELM Model and Algorithm

2.1 Problem Setup

Let $X \subseteq \mathbb{R}^d$ be the input space and $Y \subseteq \mathbb{R}$ be the output space, with unknown probability measure ρ on $Z := X \times Y$ that admits a decomposition $\rho(x, y) = \rho_X(x)\rho(y|x)$. Given a sample set $z = (x_i, y_i)_{i=1}^m$ drawn independently according to ρ, the goal is to find an estimator that minimizes the generalization error:

$$E(f) = \int_Z (f(x) - y)^2 d\rho \tag{1}$$

The hypothesis space in ELM is defined as:

$$\mathcal{M}_n = \left\{ f_n(\alpha, \beta, x) = \sum_{i=1}^{n} \alpha_i \phi(\beta_i, x) : n \in \mathbb{N} \right\} \tag{2}$$

where $x \in X$, $\beta = (\beta_1, \beta_2, \ldots, \beta_n)^T \in \mathbb{R}^{n \times l}$ with $\beta_i \in \mathbb{R}^l$, $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n)^T \in \mathbb{R}^n$, and $\phi : \mathbb{R}^l \times \mathbb{R}^n \to \mathbb{R}$ is a nonlinear activation function.

Algorithm 1 Standard ELM Algorithm

1: **Given:** Training samples $z = (x_i, y_i)_{i=1}^m$, nonlinear function ϕ, and hidden neuron number n
2: Randomly assign hidden parameters $\tilde{\beta}_i$ in $\mathbb{R}^l, i = 1, \ldots, n$
3: Calculate the hidden layer output matrix H, where:

$$H = \begin{bmatrix} \phi(\tilde{\beta}_1, x_1) & \phi(\tilde{\beta}_2, x_1) & \cdots & \phi(\tilde{\beta}_n, x_1) \\ \phi(\tilde{\beta}_1, x_2) & \phi(\tilde{\beta}_2, x_2) & \cdots & \phi(\tilde{\beta}_n, x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi(\tilde{\beta}_1, x_m) & \phi(\tilde{\beta}_2, x_m) & \cdots & \phi(\tilde{\beta}_n, x_m) \end{bmatrix}$$

4: Calculate the output weight vector $\alpha^* = H^\dagger y$ where $H^\dagger$ is a Moore-Penrose generalized inverse of H and $y = (y_1, y_2, \ldots, y_m)^T$

2.2 ELM Algorithm

The ELM algorithm differs from traditional FNN training in that it only adjusts the output weights while randomly fixing the hidden layer parameters. The algorithm consists of the following steps:

The ELM estimator is then defined as:

$$f_{ELM}(x) = f_n(\alpha^*, \tilde{\beta}, x) = \sum_{i=1}^{n} \alpha_i^* \phi(\tilde{\beta}_i, x) \tag{3}$$

This approach reduces the complex nonlinear optimization problem of traditional FNN training to a simple linear least squares problem, significantly reducing computational complexity.

3 Theoretical Findings on Generalization Capability

3.1 Baseline for Comparison

To assess whether ELM degrades generalization capability, the paper first establishes a baseline of FNN estimators. For functions with smoothness order r, the optimal generalization bound is known to be:

$$C_1 m^{-\frac{2r}{2r+d}} \leq e_m(\Psi_1) \leq C_2 m^{-\frac{2r}{2r+d}} \log m \tag{4}$$

where m is the number of samples, d is the input dimension, and r is the smoothness order of the target function.

3.2 ELM with Polynomial Activation Functions

For the case where $\phi(\beta_i, x) = (\omega_i x + b_i)^s$ with $\beta_i = (\omega_i, b_i) \in [0,1]^{d+1}$, both ω_i and b_i drawn independently according to uniform distribution in $[0,1]^d$ and $[0,1]$, and $s \in \mathbb{N}$, the paper proves:

Theorem 1: Assume $f_\rho \in F(r, C_0)$ with $0 < r \leq 1$ and $C_0 \geq 0$, and ρ_X is absolutely continuous with respect to Lebesgue measure on X. Let f_{ELM_P} be the ELM estimator. Then, whenever $s = [m^{1/(d+2r)}]$ and $n = [m^{d/(d+2r)}]$, there holds:

$$C_1 m^{-\frac{2r}{d+2r}} \leq \mathbb{E}_{\rho^m} \mathbb{E}_{\mu^n} \left(\|\pi_M(f_{ELM_P}) - f_\rho\|_\rho^2 \right) \leq C_2 m^{-\frac{2r}{d+2r}} \log m \tag{5}$$

where π_M is a truncation operator and C_1, C_2 are positive constants.

The proof demonstrates that for any $n \geq \dim(P_s^d)$, the hypothesis space of ELM with polynomial activation equals the space of all polynomials of degree at most s with d variables. This means ELM with polynomial activation does not degrade the generalization capability of FNNs.

3.3 ELM with Nadaraya-Watson Activation Functions

For Nadaraya-Watson activation functions defined as:

$$\phi_{NW}(t_i, x) = \frac{e^{-A\|x-t_i\|}}{\sum_{j=1}^{n} e^{-A\|x-t_j\|}} \tag{6}$$

where $A > 0$ is the width parameter, the paper establishes:

Theorem 2: Let $0 < r \leq 1, C_0 \geq 0$ and $f_{ELM_{NW}}$ be the ELM estimator with Nadaraya-Watson activation. If $f_\rho \in F(r, C_0)$, $n = [m^{d/(d+2r)}]$, and $A \geq n^{1/d} \log n$, then:

$$C_1 m^{-\frac{2r}{2r+d}} \leq \mathbb{E}_{\rho^m} \mathbb{E}_{\mu^n} \left(\|\pi_M(f_{ELM_{NW}}) - f_\rho\|_\rho^2 \right) \leq C_2 m^{-\frac{2r}{2r+d}} \log m \tag{7}$$

3.4 ELM with Sigmoid Activation Functions

For bounded sigmoid activation functions $\phi(\beta_i, x) = \sigma(b(x - t_i))$ with $\beta_i = (b, t_i)$, where σ is a bounded sigmoid function, $b > 0$, and t_i drawn according to uniform distribution, the paper proves:

Theorem 3: Let $0 < r \leq 1, f_\rho \in F(r, C_0)$ with $C_0 > 0$, and f_{ELM_S} be the ELM estimator with sigmoid activation. If $d = 1$, b satisfies $b \geq n h_D^{-1}$ (where h_D is the mesh norm), and $n = [m^{1/(1+2r)}]$, then:

$$C_1 m^{-\frac{2r}{2r+1}} \leq \mathbb{E}_{\rho^m} \mathbb{E}_{\mu^n} \left(\|\pi_M(f_{ELM_S}) - f_\rho\|_\rho^2 \right) \leq C_2 m^{-\frac{2r}{2r+1}} \log m \tag{8}$$

3.5 Implications of Generalization Results

These theoretical findings demonstrate that ELM with appropriate activation functions can achieve the optimal generalization bound $O(m^{-2r/(2r+d)})$. This means ELM does not degrade the generalization capability of FNNs even when connections with and within hidden neurons are randomly fixed.

A key observation is that the number of hidden neurons n is closely related to the number of training samples m. For instance, when using polynomial activation functions, the optimal number of hidden neurons is $n = [m^{d/(d+2r)}]$.

It's important to note that the generalization capability is measured in terms of double expectations $\mathbb{E}_{\rho^m} \mathbb{E}_{\mu^n}$. This indicates that on average (over all possible random assignments of hidden parameters), ELM performs as well as traditional FNNs, though individual implementations might vary.

4 Guaranteeing Weak Regularity of Hidden Layer Output Matrix

The paper addresses how to ensure the weak regularity of the hidden layer output matrix H, which is necessary for efficient application of the generalized inverse technique.

4.1 Polynomial Activation and Weak Regularity

Theorem 4: Assume the marginal distribution ρ_X is absolutely continuous with respect to Lebesgue measure on X, and algebraic polynomial is used as the activation function, i.e., $\phi(\beta_i, x) = (\omega_i x + b_i)^s$. If $n \leq m$, then the hidden layer output matrix $H = (r_{ji})_{m\times n}$ is of full column rank almost surely, where $r_{ji} = (w_i x_j + b_i)^s$.

This theorem guarantees that when using polynomial activation functions and ensuring the number of hidden neurons does not exceed the number of samples, the Moore-Penrose generalized inverse can be directly computed as $H^\dagger = (H^T H)^{-1} H^T$.

4.2 Regularization for Non-Polynomial Activations

For non-polynomial activation functions, the paper proposes using Tikhonov regularization to guarantee weak regularity without sacrificing generalization capability. The regularized ELM estimator is defined as:

$$f_{ELM}^{(\lambda)}(x) = \sum_{i=1}^{n} \alpha_i^* \phi(\tilde{\beta}_i, x) \tag{9}$$

where α^* is the solution to:

$$\alpha^* = \arg\min_{\alpha} \left\{ \frac{1}{m} \|H\alpha - y\|_2^2 + \lambda \|\alpha\|_2^2 \right\} = (H^T H + \lambda m I)^{-1} H^T y \tag{10}$$

Here, $\lambda := \lambda(m, n)$ is a regularization parameter.

Theorem 5: Let $0 < r \leq 1, C_0 > 0, f_\rho \in F(r, C_0)$ and $f_{ELM}^{(\lambda)}$ be the regularized ELM estimator. Then:

$$\mathbb{E}_{\rho^m}\mathbb{E}_{\mu^n}\left(\|\pi_M f_{ELM}^{(\lambda)} - f_\rho\|_\rho^2 \right) \leq C \frac{n(\log m - \log \lambda)}{m} + \inf_{f\in\mathcal{M}_n} \left\{ \int_X (f(x) - f_\rho(x))^2 d\rho + \lambda \|f\|_{l^2}^2 \right\} \tag{11}$$

For specific choices of activation functions and regularization parameters, the paper shows that regularized ELM maintains optimal generalization capability. For instance:

Corollary 1: If Nadaraya-Watson activation is used with $\lambda = [m^{-1}]$ and $n = [m^{d/(d+2r)}]$, then:

$$C_1 m^{-\frac{2r}{2r+d}} \leq \mathbb{E}_{\rho^m}\mathbb{E}_{\mu^n}\left(\|\pi_M (f_{ELM_{NW}}^{(\lambda)}) - f_\rho\|_\rho^2 \right) \leq C_2 m^{-\frac{2r}{2r+d}} \log m \tag{12}$$

Corollary 2: If sigmoid activation is used with $\lambda = [m^{-1}]$ and $n = [m^{1/(1+2r)}]$, then:

$$C_1 m^{-\frac{2r}{2r+1}} \leq \mathbb{E}_{\rho^m}\mathbb{E}_{\mu^n}\left(\|\pi_M (f_{ELM_S}^{(\lambda)}) - f_\rho\|_\rho^2 \right) \leq C_2 m^{-\frac{2r}{2r+1}} \log m \tag{13}$$

5 Numerical Examples and Results

The paper includes experimental verification of the theoretical findings using three UCI benchmark datasets: Abalone, Machine CPU, and Census (house8L). Three ELM variants were compared:

- ELM: Standard ELM with sigmoid activation function
- L2ELM: Regularized ELM with sigmoid activation function
- PolyELM: ELM with polynomial activation function

The experiments showed that:

- Both L2ELM and PolyELM exhibited stable training performance with increasing number of hidden neurons, while standard ELM with sigmoid activation became unstable.
- For testing performance (generalization), L2ELM and PolyELM showed consistently good results and remained stable as the number of hidden neurons increased. Standard ELM performed well only when the number of hidden neurons was within a suitable range and degraded beyond that range.
- The experiments confirmed that either using polynomial activation functions or employing regularization can effectively guarantee both the weak regularity of the hidden layer output matrix and good generalization performance.

These experimental results validate the theoretical findings that ELM with appropriate activation functions or regularization does not degrade generalization capability compared to traditional FNNs.

6 Implications

The theoretical assessment in this paper has several important implications for the use of ELM in practice:

- **Proper activation selection:** The choice of activation function significantly affects ELM performance. Polynomial, Nadaraya-Watson, and sigmoid activations are proven to maintain optimal generalization capability.
- **Hidden neuron sizing:** The paper establishes a precise relationship between the optimal number of hidden neurons and the number of training samples. This relationship depends on the dimensionality of the input space and the smoothness of the target function.
- **Regularization benefits:** For non-polynomial activation functions, Tikhonov regularization effectively addresses the weak regularity issue without sacrificing generalization capability.
- **Computational advantage:** ELM offers significant complexity reduction compared to traditional FNNs by only training output weights while maintaining generalization performance, making it suitable for large-scale applications.
- **Theoretical justification:** The paper provides the first comprehensive theoretical justification for why ELM works in practice, explaining the surprising effectiveness of random feature mapping.

The analysis reveals that ELM's feasibility depends on both appropriate activation function selection and proper configuration of the number of hidden neurons relative to training samples. When properly configured, ELM achieves optimal generalization bounds comparable to traditional FNNs while maintaining significantly lower computational complexity.